Dear Quillstone Payments Hiring Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. As a Data Scientist at Thistlebox, I have built forecasting, churn-scoring and experimentation solutions using Python, SQL and managed GCP services including BigQuery and Vertex AI Workbench. I am keen to bring this practical experience in model development, evaluation and scheduled ML workflows to a team building AI systems for merchants and support teams, while continuing to develop my production machine-learning engineering skills.

I developed a weekly recipe-demand forecasting workflow that produces six-week-ahead forecasts by recipe and region. It runs as a scheduled notebook in Vertex AI Workbench and writes results to BigQuery for operational use in Looker. The work improved WAPE from 23% to 14% and was associated with an estimated 9% reduction in food waste. I have also built a weekly XGBoost churn-scoring workflow in BigQuery; in a 10% holdout evaluation, churn was 2.1 percentage points lower in the treated group.

Alongside model development, feature engineering and scheduled data workflows, I built a Streamlit application for planning and analysing sequential A/B tests, used by around 25 colleagues in product and growth, and wrote its Dockerfile. The data platform team handled deployment to Cloud Run. I have also begun writing pytest unit tests for forecasting feature code, with coverage now at approximately 60%. My MSc in Statistics, including a dissertation on Bayesian hierarchical models, underpins my approach to modelling and evaluation, and I am motivated to continue building my engineering skills as part of a collaborative team.

I would welcome the opportunity to discuss how my experience developing and evaluating Python-based machine-learning workflows, including scheduled forecasting, batch churn scoring and automated testing, could contribute to Quillstone Payments' Customer Operations AI team. Thank you for considering my application.

Kind regards,
Tobias Mwangi-Hartley